

Deep Early-Exercise Premium Physics-Informed Neural Networks (Deep-EEP-PINN): High-Dimensional American Basket Option Free-Boundary Valuation up to 50 Dimensions

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Abstract

Valuing high-dimensional American basket options represents a challenging parabolic free-boundary obstacle partial differential equation (PDE) problem. Conventional grid-based solvers (e.g., Finite Difference, Projected Successive Over-Relaxation) suffer from Bellman’s curse of dimensionality, while standard Physics-Informed Neural Networks (PINNs) fail due to the non-differentiable payoff kink singularity at expiration $t = T$ and Dirac-delta blow-up of spatial Gamma (Γ). In this paper, we introduce **Deep-EEP-PINN**, a mesh-free scientific machine learning architecture for multi-asset American option pricing up to $d = 50$ correlated assets (1,225 pairwise correlation terms). Deep-EEP-PINN combines the analytical Kim (1990) / Carr-Jarrow-Myneni (1992) Early Exercise Premium (EEP) decomposition with a hard-constraint boundary-encoded neural ansatz that identically satisfies $e_\theta(\mathbf{S}, T) \equiv 0$, eliminating the terminal kink singularity by construction. For multi-asset cross-diffusion, we introduce an $\mathcal{O}(d)$ directional autograd Hessian trace contraction combined with chunked loss backpropagation, constraining peak GPU VRAM to < 3 GB on standard accelerators. Rigorous empirical validation against 100,000-path Longstaff-Schwartz Least Squares Monte Carlo (LSM) demonstrates that Deep-EEP-PINN achieves mean pricing errors of Rs. 0.06 (6 paise) in 30D (Dow Jones scale) and Rs. 0.08 (8 paise) in 50D (Nifty 50 scale) with an inference latency of 3.94 ms, delivering a **176,760 $\times$** speedup over 100k-path LSM regressions.

Keywords: Physics-Informed Neural Networks (PINNs), Free-Boundary PDEs, American Basket Options, Optimal Stopping, High-Dimensional Scientific Machine Learning, Curse of Dimensionality.

1 Introduction

American-style financial derivatives permit the holder to exercise the contract early at any stopping time $\tau \in [0, T]$ prior to maturity T . The continuous-time valuation of multi-asset American basket options under correlated Geometric Brownian Motion constitutes a parabolic obstacle problem governed by a Linear Complementarity Problem (LCP):

$$\min \left(-\frac{\partial V}{\partial t} - \mathcal{L}_{\text{BS}} V, V(\mathbf{S}, t) - h(\mathbf{S}) \right) = 0, \quad (\mathbf{S}, t) \in \mathbb{R}_+^d \times [0, T), \quad (1)$$

subject to the terminal obstacle condition $V(\mathbf{S}, T) = h(\mathbf{S})$, where $h(\mathbf{S}) = \max(K - \phi(\mathbf{S}), 0)$ is the intrinsic payoff, $\phi(\mathbf{S})$ is the basket aggregation function, and $\mathcal{L}_{\text{BS}}$ denotes the multi-asset

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Black-Scholes spatial diffusion operator:

$$\mathcal{L}_{\text{BS}}V = \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V}{\partial S_i \partial S_j} + r \sum_{i=1}^d S_i \frac{\partial V}{\partial S_i} - rV. \quad (2)$$

In one spatial dimension ($d = 1$), classical numerical methods such as Projected Successive Over-Relaxation (PSOR) and Crank-Nicolson finite difference schemes achieve high precision [8, 9]. However, as the asset dimension increases ($d \geq 5$), grid-based discretization collapses due to the curse of dimensionality, requiring $\mathcal{O}(N^d)$ spatial nodes.

While recent deep learning approaches based on Backward Stochastic Differential Equations (BSDEs) and deep optimal stopping [5–7] offer path-wise approximations, they remain simulation-dependent, require thousands of Monte Carlo samples per evaluation point, and do not provide continuous closed-form spatial derivatives (Greeks) via automatic differentiation.

Conversely, standard Physics-Informed Neural Networks (PINNs) [3, 12] suffer from severe optimization stalling and Dirac-delta Gamma blow-up at expiration due to the non-smooth kink in $\max(K - S, 0)$.

2 Methodology: The Deep-EEP-PINN Architecture

2.1 Analytical Early Exercise Premium Decomposition

Following the analytical foundations established by Kim [1] and Carr, Jarrow, and Myneni [2], the total American option value is decomposed into an exact European anchor and an early exercise premium:

$$V_{\text{Amer}}(\mathbf{S}, t) = V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) + e_{\theta}(\mathbf{S}, t), \quad (3)$$

where $V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t)$ is evaluated via exact closed-form Black-Scholes formulas (or lognormal moment-matched approximations for arithmetic baskets [10, 11]).

2.2 Hard-Constraint Boundary-Encoded Neural Ansatz

To guarantee $e_{\theta}(\mathbf{S}, T) \equiv 0$ strictly without training error, the premium is parameterized as:

$$e_{\theta}(\mathbf{S}, t) = \text{Softplus}(\mathcal{N}_{\theta}(\mathbf{S}, t)) \cdot \left(\frac{T-t}{T} \right) \cdot \prod_{i=1}^d \left(1 - \frac{S_i}{S_{\max}} \right) \cdot K (1 - e^{-rT}). \quad (4)$$

2.3 $\mathcal{O}(d)$ Directional Autograd Trace Contraction

To evaluate the cross-diffusion term $\frac{1}{2} \text{Tr}(\mathbf{\Sigma}(\mathbf{S})\mathbf{H})$ without constructing the $d \times d$ Hessian matrix, we decompose the covariance matrix into its Cholesky factor $\mathbf{L}$ ($\mathbf{\Sigma} = \mathbf{L}\mathbf{L}^T$) and compute directional second derivatives:

$$\frac{1}{2} \sum_{k=1}^d \mathbf{v}_k^T \nabla_{\mathbf{S}}^2 e_{\theta} \mathbf{v}_k, \quad \mathbf{v}_k = \mathbf{L}_k \odot \mathbf{S}. \quad (5)$$

3 Empirical Results & Benchmark Suite

All experiments were executed on an NVIDIA Tesla P100 GPU (16GB VRAM) and validated against 100,000-path Longstaff-Schwartz Least Squares Monte Carlo regressions.

Table 1: Master Benchmark Results across All Dimensions ($d = 1, 5, 10, 30, 50$).

Problem Setup	Ground Truth	EEP-PINN Error	Inference Latency	Spe
1D American Put ($d = 1$)	PSOR (90k nodes)	$L_2 = 0.37\%$, MAE = Rs. 0.08	11.84 ms	97
5D Geometric Basket ($d = 5$)	100k LSM ($\pm 1.96\text{SE}$)	Diff = Rs. 0.06 (ATM)	3.50 ms	27
5D Arithmetic Basket ($d = 5$)	100k LSM ($\pm 1.96\text{SE}$)	Diff = Rs. 0.02 (ATM)	7.55 ms	10
10D High-Dim Basket ($d = 10$)	100k LSM ($\pm 1.96\text{SE}$)	Diff = Rs. 0.14 (Mean)	3.47 ms	68
30D Dow Jones Scale ($d = 30$)	100k LSM (498 basis)	Diff = Rs. 0.06 (Mean)	3.26 ms	62,
50D Nifty 50 Scale ($d = 50$)	100k LSM (1328 basis)	Diff = Rs. 0.08 (Mean)	3.94 ms	176,

4 Conclusion

Deep-EEP-PINN establishes a scalable, mesh-free framework for continuous-time free-boundary obstacle PDEs in mathematical finance, breaking the curse of dimensionality up to 50 correlated assets while providing real-time millisecond inference for high-frequency risk management.

References

- [1] I. J. Kim, *The Analytic Valuation of American Options*, The Review of Financial Studies, 3(4):547–572, 1990.
- [2] P. Carr, R. Jarrow, and R. Myneni, *Alternative Methods for Valuing American Options*, Finance and Stochastics, 2(1):87–106, 1992.
- [3] M. Raissi, P. Perdikaris, and G. E. Karniadakis, *Physics-informed neural networks*, Journal of Computational Physics, 378:686–707, 2019.
- [4] F. A. Longstaff and E. S. Schwartz, *Valuing American Options by Simulation: A Simple Least-Squares Approach*, The Review of Financial Studies, 14(1):113–147, 2001.
- [5] J. Han, A. Jentzen, and W. E, *Solving high-dimensional partial differential equations using deep learning*, PNAS, 115(34):8505–8510, 2018.
- [6] S. Becker, P. Cheridito, and A. Jentzen, *Deep Optimal Stopping*, Journal of Machine Learning Research, 20(74):1–25, 2019.
- [7] S. Becker, P. Cheridito, A. Jentzen, and T. Welti, *Solving High-Dimensional Optimal Stopping Problems Using Deep Learning*, SIAM J. Sci. Comput., 42(1):B1–B29, 2020.
- [8] C. W. Cryer, *The Solution of a Quadratic Programming Problem using Systematic Overrelaxation*, SIAM J. Control, 9(3):385–392, 1971.
- [9] M. J. Brennan and E. S. Schwartz, *The Valuation of American Put Options*, The Journal of Finance, 32(2):449–462, 1977.
- [10] M. A. Milevsky and S. E. Posner, *Valuing Exotic Options by Approximating the Swapping Density*, JFQA, 33(3):409–425, 1998.
- [11] D. Gentle, *Basket Options*, Risk, 6(12):51–52, 1993.
- [12] A. Al-Aradi, A. Correia, D. de Freitas, and P. Garnier, *Solving nonlinear and high-dimensional PDEs in finance using PINNs*, arXiv:1811.08782, 2018.